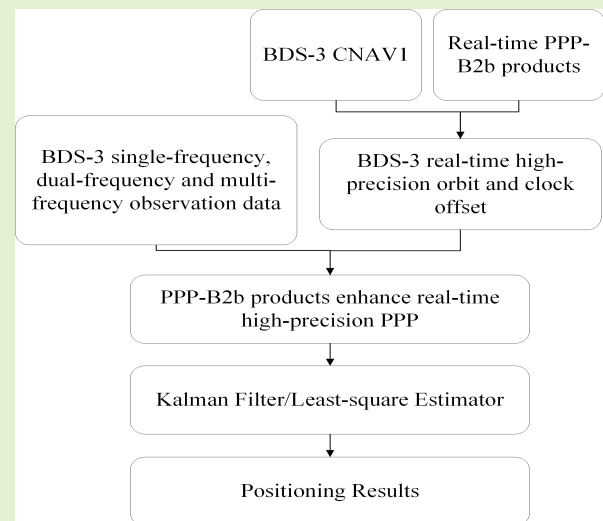


Multi-Frequency BDS-3 Real-Time Positioning Performance Assessment Using New PPP-B2b Augmentation Service

Haitao Zhou¹, Wenju Fu¹, Lei Wang¹, Tao Li¹, Yuan Wu¹, Ruizhi Chen¹, and Juanjuan Li¹

Abstract—A new precise point positioning (PPP)-B2b augmentation service broadcast by the BeiDou Navigation Satellite System (BDS-3) geosynchronous Earth orbit (GEO) satellite can provide real-time and high-precision orbit and clock offset corrections for global navigation satellite system (GNSS) users in China and its surrounding areas. It has great significance and research value for real-time and high-precision positioning applications. First, orbit, clock offset, and differential code bias (DCB) of PPP-B2b products are evaluated. Second, the influence of PPP-B2b service on the enhanced positioning of single-frequency (SF), dual-frequency (DF), and Multi-Frequency (MF) real-time PPP using 30 days of BDS-3 observations is verified. The result shows that the standard deviation (STD) of clock offset and the root mean square (rms) of orbit in the 3-D direction for medium Earth orbit (MEO) satellites are 0.118 ns and 0.286 m, respectively, which is 75.9% and 18.3% higher than that of broadcast ephemeris. The statistical results show that the median positioning accuracy of static SF PPP, DF PPP, and MF PPP is better than 0.20/0.09/0.08 m, and the convergence time is better than 51/10/8 min. The median positioning accuracy of kinematic SF PPP, DF PPP, and MF PPP is better than 0.40/0.12/0.12 m, and the convergence time is better than 145/16/12 min. Using the PPP-B2b products, the positioning accuracy of DF PPP and MF PPP is comparable and close to that of DF PPP using the precise products, while the convergence time of MF PPP is improved by 24.8% and 27.7% over DF PPP in static and kinematic solutions, respectively.

Index Terms—BeiDou Navigation Satellite System (BDS-3), multi-frequency (MF), PPP-B2b service, precise point positioning (PPP), real time.



I. INTRODUCTION

THE BeiDou Navigation Satellite System (BDS-3) is one of the four major navigation and positioning systems in the world, which is independently designed, developed, and operated by China. On June 23, 2020, the deployment of

BDS-3 constellation was completed, indicating that BDS-3 can provide navigation, positioning, and timing (PNT) services to global users [2]. The BDS-3 includes 24 medium Earth orbit (MEO) satellites, three inclined geosynchronous satellite orbit (IGSO) satellites and three geosynchronous Earth orbit (GEO) satellites [3], [4]. BDS-3 GEO satellite-based regional featured services consist of the radio determination satellite service (RDSS), the BDS satellite-based augmented service (BDS-BAS), the regional short message communication (RSMC) services, and the satellite-based precise point positioning (PPP) service via B2b signal (PPP-B2b) [2], [5], [6].

The PPP technology uses a single global navigation satellite system (GNSS) receiver to achieve high-precision positioning by receiving pseudorange and phase observations and using high-precision satellite orbit and clock offset products. It has important applications in low-orbit satellite orbit determination [7], [8], water vapor inversion [9], seismic monitoring,

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ionospheric monitoring [10], [11], and other fields. Since PPP was put forward in 1997, it has gone through the development process from ambiguity float solution to fixed solution [12], [13], [14] and from postprocessing to real-time processing [15]. The research of PPP mainly focuses on the PPP positioning model, data preprocessing, error model refinement, ambiguity fixing, and fast initialization [16]. The emergence of the Multi-Frequency (MF) GNSS technology [17], [18] can provide more redundant observations for PPP solution and can also construct more optimal combinations [19] to meet the purpose of cycle slip detection [20] or ambiguity resolution, and finally improve the positioning accuracy and speed up convergence.

With the rapid development of GNSS technology, real-time PPP [21], [22] has become a research hotspot in the GNSS community, which has significant application prospects in the field of unmanned driving, smart agriculture, and earthquake warning. However, the main constraint to real-time PPP is the availability of the GNSS orbit and clock offset products with high accuracy in real time [23], [24], [25]. In 2013, IGS organizations began to broadcast real-time precise orbit and clock offset data streams to global users via network according to the standard of Radio Technical Commission for Maritime Service (RTCM). Centre National D'Etudes Spatiales (CNES) and other organizations also broadcast real-time corrections of the multi-GNSS systems to global users [26]. The positioning accuracy of static centimeter level and kinematic decimeter level can be achieved. However, PPP may be reinitialized when the network is interrupted. In addition, the PPP-B2b service broadcast by BDS-3 GEO satellite via the B2b signal has also started to broadcast the BDS-3/GPS precise orbit and clock offset corrections to China and its surrounding areas since 2020 [27]. At present, some studies have been conducted to enhance dual-frequency (DF) PPP based on PPP-B2b corrections [28], [29], [30], [31] and apply it to time-frequency transmission and earthquake warning [32], [33], but the positioning performance of single-frequency (SF) and MF PPP has not been studied.

The main contribution of this article is to comprehensively assess the quality of PPP-B2b products and to analyze the enhanced positioning effect of PPP-B2b products on SF, DF, and MF PPP from the perspective of positioning accuracy and convergence time, which can provide more options for PPP users. This article is organized as follows. First, this article introduces the assessment methods of PPP-B2b products and briefly gives the SF, DF, and MF PPP models. Then, the accuracy of PPP-B2b products is evaluated, and the positioning performance of PPP-B2b service-enhanced BDS-3 SF, DF, and MF PPP is evaluated. Finally, some conclusions and findings are summarized.

II. MF PPP MODEL WITH PPP-B2B CORRECTIONS

Currently, BDS-3 broadcasts only four types of PPP-B2b corrections through GEO satellites (C59, C60, and C61). Type 1 broadcasts the pseudo-random noise (PRN) mask of BDS-3 satellites, and the PRN mask of the satellites with PPP-B2b corrections is 1; otherwise, it is 0. Type 2 broadcasts issue of data, navigation (IODN), IOD Corr, orbit corrections, and

corresponding user range accuracy index (URI). Type 3 broadcasts differential code bias (DCB). Type 4 broadcasts IOD Corr and clock offset corrections. The update time is 6 s for Type 4 and 48 s for the rest of the types. The PPP-B2b corrections can be combined with the broadcast ephemeris (BRDC) to recover the real-time orbit and clock offset with high accuracy. This section briefly introduces the evaluation method of PPP-B2b corrections and three MF BDS-3 PPP models.

A. PPP-B2b Corrections

The orbit and clock offset of PPP-B2b corrections must be matched to be used. If IOD Corr in Type 2 matches that in Type 4, the IODN in Type 2 must also match the issue of data, clock (IODC) in the broadcast ephemeris. Then, the PPP-B2b corrections can correct the orbit in radial, along, and cross (RAC) directions and the clock offset of the broadcast ephemeris. The reference points of orbit and clock offset of broadcast ephemeris and PPP-B2b products are the B3I frequency center of BDS-3 satellites. For final precise multi-GNSS products from Geo Forschungs Zentrum (GFZ) products, the orbit reference point of BDS-3 satellites is the satellite center-of-mass (CoM), while the clock offset reference point is the B3I frequency center. When evaluating the accuracy of orbit and clock offset of PPP-B2b products, the corresponding conversion should be carried out for comparison [6], [28].

The evaluation of clock offset accuracy is relatively complicated. Clock offset provided by different analysis centers cannot be compared directly due to the differences in the underlying GNSS-specific system time scales [34]. There is a systematic bias after the single-difference (SD) value is formed between the PPP-B2b products and the GFZ products. In this case, the systematic bias of the satellite can be eliminated by subtracting the SD value of the reference satellite with a longer observation arc or correcting an average ensemble clock difference computed at each epoch from the SD values of satellites in a constellation [6]. This article adopts the second way to eliminate the systematic bias, and the double-difference (DD) values can be written as follows:

$$\begin{aligned} \text{DDclk}^s(i) &= \text{clk}_{\text{B2b}}^s(i) - \text{clk}_{\text{GFZ}}^s(i) \\ &\quad - \frac{1}{ns} \sum_{n=1}^{ns} (\text{clk}_{\text{B2b}}^n(i) - \text{clk}_{\text{GFZ}}^n(i)) \end{aligned} \quad (1)$$

where B2b and GFZ represent the PPP-B2b products and the final GFZ products, respectively; s represents the satellite PRN of BDS-3. clk stands for satellite clock offset; i indicates the current epoch. ns indicates the total number of available satellites in the current epoch.

The signal-in space range error (SISRE) is usually used to comprehensively reflect the impact of orbit and clock offset on final positioning. The SISRE is calculated as follows [34]:

$$\begin{aligned} \text{SISRE} &= \begin{cases} \sqrt{(0.98 \cdot R)^2 + (A^2 + C^2) / 54}, \\ \sqrt{(0.99 \cdot R)^2 + (A^2 + C^2) / 126}, \text{ BDS-3 (IGSO)} \end{cases} \end{aligned} \quad (2)$$

where R , A , and C denote the root mean square (rms) of orbit errors in RAC directions, respectively.

B. PPP Model With Different Frequencies

The PPP model uses the pseudorange and phase observation to construct the observation equations. The observation equations can be expressed as follows:

$$\begin{cases} P_r^s = \rho_r^s + dt_r - dt^s + Mw_r^s \cdot ZWD_r + I_r^s + \varepsilon_{r,P}^s \\ L_r^s = \rho_r^s + dt_r - dt^s + Mw_r^s \cdot ZWD_r - I_r^s + \lambda_i^s N_i^s + \varepsilon_{r,L}^s \end{cases} \quad (3)$$

where s and r represent the satellite PRN and the receiver, respectively; P and L denote the code and phase observations, respectively; ρ is the geometric range between the satellite and receiver antennas; dt_r and dt^s denote the receiver and satellite clock offsets, respectively; I is slant ionospheric delay; Mw_r^s and ZWD_r are the mapping function of zenith tropospheric wet delay and corresponding zenith troposphere wet delay, respectively; λ_i and N_i^s are the wavelength of different frequencies and the ambiguity of the specific satellite in cycles, respectively; $\varepsilon_{r,P}^s$ and $\varepsilon_{r,L}^s$ are the noise of code and phase observations. Unless otherwise stated, the units used in this article are in meters.

1) *Single-Frequency PPP Model*: The ionosphere effect is a key issue for SF PPP. In contrast to the conventional SF PPP to eliminate the ionosphere by employing code-minus-phase combination, we estimate the ionosphere delay as parameter in this study [1]. Ignoring second-order ionospheric delay [35], the observation equation of the SF PPP model is shown as follows:

$$\begin{cases} \text{OMC}(P_r^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r + I_r^s + \varepsilon_{r,P}^s \\ \text{OMC}(L_r^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r - I_r^s + \lambda_i^s N_i^s + \varepsilon_{r,L}^s \end{cases} \quad (4)$$

where OMC represents the observed value minus the calculated value. Pseudorange or phase observations subtract the errors calculated by the error models. The relevant error models include tropospheric dry delay, phase windup, phase center offset (PCO) correction, relativistic correction, tidal correction, and so on. x_r^s is the coordinate parameter. Therefore, after deducting the related errors, the parameters of SF PPP include receiver position, receiver clock offset, tropospheric wet delay, ionospheric delay, and ambiguity. When there are n visible satellites, the parameters of SF PPP can be expressed as follows:

$$X = [x_{3 \times 1} \ dt_r \ ZWD \ I_{n \times 1} \ N_{n \times 1}] \quad (5)$$

In addition, B1I observations are used for the SF PPP model in this study, and the frequency reference of orbit and clock offset needs to be converted to B1I signal. The orbit can be corrected by PCO, and the clock offset can be corrected by DCB. The clock offset correction formula is as follows:

$$\bar{T}_{\text{broadcast_C}}^{\text{B1I}} = \hat{T}_{\text{broadcast_C}} + \text{DCB}_{\text{B1I-B3I}} \quad (6)$$

$$\bar{T}_{\text{precise_C}}^{\text{B1I}} = T_{\text{precise_C}} - \frac{1}{\beta - 1} \text{DCB}_{\text{B1I-B3I}} \quad (7)$$

where $\beta = f_{\text{B1I}}^2 / f_{\text{B3I}}^2$; $\bar{T}_{\text{broadcast_C}}^{\text{B1I}}$ and $\bar{T}_{\text{precise_C}}^{\text{B1I}}$ are the clock offset considering DCB; $\hat{T}_{\text{broadcast_C}}$ and $T_{\text{precise_C}}$ are BDS-3

satellites clock offset for broadcast ephemeris and final GFZ products, respectively. $\text{DCB}_{\text{B1I-B3I}}$ is the DCB obtained from PPP-B2b products.

2) *DF PPP Model*: The DF PPP model adopts an ionosphere-free (IF) combination, and the DF observation equations can be expressed as follows:

$$\begin{cases} \text{OMC}(P_{r,\text{IF}jk}^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r + \varepsilon_{r,P,\text{IF}jk}^s \\ \text{OMC}(L_{r,\text{IF}jk}^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r \\ \quad + \lambda_{i,\text{IF}jk}^s N_{i,\text{IF}jk}^s + \varepsilon_{r,L,\text{IF}jk}^s \end{cases} \quad (8)$$

where j and k represent the B1I and B3I signal frequencies; $P_{r,\text{IF}jk}^s$ and $L_{r,\text{IF}jk}^s$ represent the IF combination of pseudorange and phase; $\lambda_{i,\text{IF}jk}^s$ and $N_{i,\text{IF}jk}^s$ represent the wavelength and ambiguity of IF combination for special satellite; $\varepsilon_{r,P,\text{IF}jk}^s$ and $\varepsilon_{r,L,\text{IF}jk}^s$ are the measurement noises of pseudorange and phase IF combinations. Other parameters are the same as above. B1I and B3I observations of BDS-3 satellites are adopted to construct the DF PPP model in this study. The DCB should be considered for B1I/B3I IF combination [36], and the formula is as follows [28]:

$$\begin{aligned} d_{\text{IF}ij}^s - d_{\text{B3I}}^s &= \frac{f_j^2}{f_j^2 - f_k^2} (d_j^s - d_{\text{B3I}}^s) - \frac{f_k^2}{f_j^2 - f_k^2} (d_k^s - d_{\text{B3I}}^s) \\ &= \frac{f_j^2}{f_j^2 - f_k^2} \text{DCB}_j^s - \frac{f_k^2}{f_j^2 - f_k^2} \text{DCB}_k^s \end{aligned} \quad (9)$$

where f is the frequency; d represents the satellite code hardware delay; DCB can be obtained from PPP-B2b products.

3) *MF PPP Model*: Based on the B1I/B3I observations, BDS-3 also broadcasts B1C/B2a signals at the same time, which provides more observations for MF PPP. To verify the enhancement effect of PPP-B2b products on MF PPP, B1C/B2a IF combination is added based on the B1I/B3I IF combination. The pseudorange and phase observation equations of MF PPP are as follows:

$$\begin{cases} \text{OMC}(P_{r,\text{IF}jk}^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r + \varepsilon_{r,P,\text{IF}jk}^s \\ \text{OMC}(L_{r,\text{IF}jk}^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r \\ \quad + \lambda_{i,\text{IF}jk}^s N_{i,\text{IF}jk}^s + \varepsilon_{r,L,\text{IF}jk}^s \\ \text{OMC}(P_{r,\text{IF}ac}^s) = x_r^s + dt_r + \text{IFB} + Mw_r^s \cdot ZWD_r \\ \quad + \varepsilon_{r,P,\text{IF}ac}^s \\ \text{OMC}(L_{r,\text{IF}ac}^s) = x_r^s + dt_r + Mw_r^s \cdot ZWD_r \\ \quad + \lambda_{i,\text{IF}ac}^s N_{i,\text{IF}ac}^s + \varepsilon_{r,L,\text{IF}ac}^s \end{cases} \quad (10)$$

where a and c represent the B2a/B1C signal frequency; IFB is the interfrequency bias of receiver, and the random walk model is used to estimate this parameter. Other parameters are the same as above. The B1C/B2a IF ambiguity should also be added to parameter estimation. The parameters for MF PPP are as follows:

$$X = [x_{3 \times 1} \ dt_r \ \text{IFB} \ ZWD \ N_{jk,n \times 1} \ N_{ac,n \times 1}] \quad (11)$$

Similarly, DCB should also be considered for the B1C/B2a IF combination, and the correction method should refer to (9).

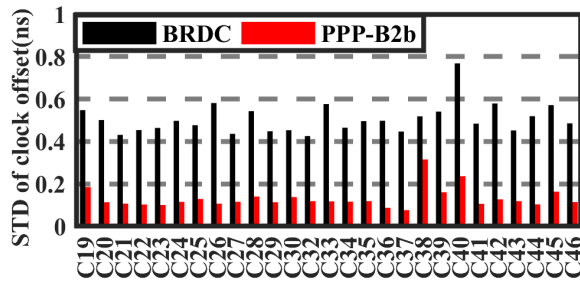


Fig. 1. Mean STD of clock offset for BDS-3 with BRDC and PPP-B2b products.

TABLE I

BDS-3 MEO SATELLITES OF DIFFERENT MANUFACTURERS

Manufacturer	MEO PRN of BDS-3
CASC	C19,C20,C21,C22,C23,C24,C32,C33,C36,C37,C41,C42,C45,C46
SECM	C25,C26,C27,C28,C29,C30,C34,C35,C43,C44

III. ACCURACY ASSESSMENT OF PPP-B2B CORRECTIONS

This section mainly evaluates the accuracy of PPP-B2b products, including orbit, clock offset, and DCBs. The dataset decoded by Septentrio R5 receiver was obtained from June 1 to 30, 2022. In addition, the orbit and clock offset results of broadcast ephemeris are taken as a reference for comparison.

A. Accuracy of Clock Offset

To evaluate the accuracy of PPP-B2b clock offset, the differences of GNSS-specific system time scales were eliminated by the DD method [6], [34], and the mean standard deviation (STD) of clock offset for each BDS-3 satellites with 30 days of data is shown in Fig. 1. As shown in the figure, except for the C40 satellite, the clock offset STD of the broadcast ephemeris for the other BDS-3 satellites fluctuates within 0.4–0.6 ns. Except for the IGSO satellites (C38, C39, and C40), the clock offset STD of PPP-B2b products for all MEO satellites is less than 0.2 ns. Compared with MEO satellites, the clock offset STD of BDS-3 IGSO satellites is generally poor. The clock offset of all satellites fluctuates more smoothly with PPP-B2b corrections, which also facilitates short-term clock offset forecast in the event of the PPP-B2b service outage.

It should be noted that the BDS-3 MEO satellites are manufactured by the China Aerospace Science and Technology Corporation (CASC) and the Shanghai Engineering Center for Microsatellites (SECM) [37], and the details are shown in Table I. After the statistics, it was found that the clock offset STD of the two types of MEO satellites was 0.1167 and 0.12 ns, respectively, with a difference of 0.003 ns, indicating that the clock offset accuracy of the MEO satellites made by CASC and SECM is approximately equivalent.

The statistical results of all BDS-3 satellites are shown in Table II. The mean STD of MEO and IGSO satellites clock offset is 0.118 and 0.237 ns, respectively, which is 75.9% and 61.8% higher than that of broadcast ephemeris. The accuracy of broadcast ephemeris clock offset has been greatly improved with PPP-B2b corrections.

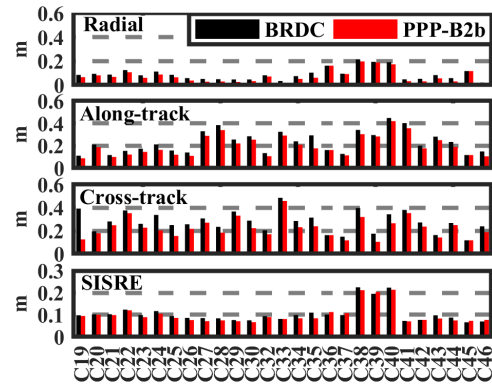


Fig. 2. Mean rms of orbit errors in (a) radial, (b) along track, (c) cross track, and (d) SISRE for BDS-3 with BRDC and PPP-B2b products.

TABLE II

MEAN RMS OF ORBIT AND STD OF CLOCK OFFSET FOR BDS-3

Satellite Type	BDS-3 MEO		BDS-3 IGSO	
Product Type	BRDC	PPP-B2b	BRDC	PPP-B2b
RMS of Orbit errors (m)	R	0.079	0.062	0.200
	A	0.209	0.175	0.362
	C	0.269	0.217	0.304
	3D	0.350	0.286	0.513
	SISRE	0.093	0.087	0.214
STD of clock offset (ns)	0.489	0.118	0.609	0.237

B. Accuracy of Orbit

The orbit accuracy of PPP-B2b products in RAC directions can be evaluated when the phase center of satellites for PPP-B2b products and final GFZ products is converted to B1I/B3I IF combination phase center of the satellite. The rms of orbit errors obtained for each satellite using 30 days of data is shown in Fig. 2. The orbit accuracy of the broadcast ephemeris is also given for comparison. The rms in the radial direction is the smallest in RAC directions and is better than 0.2 m. The rms in along-track and cross-track directions is better than 0.4 m. In general, due to the application of inter-satellite link (ISL) measurements to broadcast ephemeris, the orbit accuracy of broadcast ephemeris for BDS-3 satellites has been greatly improved [38]. Even with PPP-B2b corrections, the accuracy improvement is not obvious. At the same time, it can be seen that the orbit accuracy of C26–C30 satellites in the radial direction is very good, which may be due to more available ISL measurements of these satellites [6] during the generation of broadcast ephemeris.

The statistical results of the orbit errors of all BDS-3 satellites are shown in Fig. 2 and Table II. The orbit accuracy of the PPP-B2b products is comparable to that of the broadcast ephemeris, and the improvement is not significant over the clock offset. The rms of the PPP-B2b products of MEO and IGSO satellites in the 3-D direction is 0.286 and 0.447 m, respectively, which are 18.3% and 12.9% higher than that of the broadcast ephemeris. The SISRE of PPP-B2b products of MEO and IGSO satellites is 0.087 and 0.21 m, respectively, which are 6% and 2% higher than that of the broadcast ephemeris. PPP-B2b products cannot be greatly improved in the radial direction due to ISL measurements, but there is a large room for improvement in the along-track and cross-track

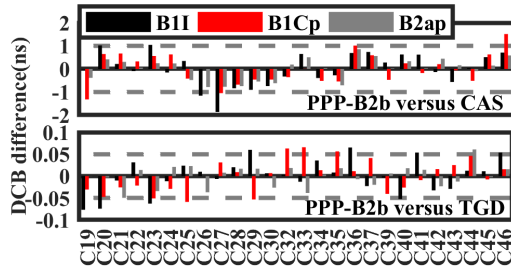


Fig. 3. Mean DCB difference of PPP-B2b products versus CAS products (top) and PPP-B2b products versus TGD (bottom).

directions. In addition, the reason why the SISRE of IGSO satellites is inferior to that of the MEO satellites may be due to the fact that this kind of regional satellites cannot be well tracked globally [2], and also due to the fact that there are fewer available ISL measurements for IGSO satellites [6].

Meanwhile, we also conducted a statistical analysis of the orbital accuracy of the BDS-3 MEO satellites manufactured by CASC and SECM. The orbit rms in the 3-D direction of the two types of the MEO satellites is 0.2838 and 0.3132 m, and the SISRE is 0.0926 and 0.0772 m, respectively. Both the orbit rms difference and the SISRE difference are about 2 cm, indicating that the orbit accuracies of BDS-3 MEO satellites manufactured by CASC and SECM are approximately comparable.

C. Accuracy of DCBs

DCB at different signal frequencies of BDS-3 satellites represents the satellite DCB between the corresponding signal frequency and the B3I signal frequency. At present, the PPP-B2b products only provide the DCBs of BDS-3 satellites, including DCB (B1I), DCB (B1Cp), and DCB (B2ap). To evaluate the DCBs accuracy of the PPP-B2b products, compare them with the multi-GNSS experiment (MGEX) DCB products provided by the Chinese Academy of Sciences (CAS) [39] and the time group delay (TGD) provided by broadcast ephemeris, respectively. The average DCB difference of the 30 days of each BDS-3 satellite is shown in Fig. 3. As can be seen from the figure, the DCB difference between the PPP-B2b products and the CAS products is relatively large, and the DCB difference of B1I/B1Cp of all satellites fluctuates between 0 and 2 ns. The DCB difference of B2ap is better than 1 ns, which was slightly better than that of B1I/B1Cp. The DCB accuracy of the PPP-B2b products and the TGD is comparable, and the difference of B1I/B1Cp/B2ap for all BDS-3 satellites is better than 0.1 ns. This may be the reason that both PPP-B2b products and TGD are provided by the China Analysis Center based on a small number of ground-based tracked stations, while CAS products are provided by solutions using global tracked stations.

IV. POSITIONING PERFORMANCE WITH PPP-B2B CORRECTIONS

A. Dataset and PPP Strategy

To verify the enhanced effect of PPP-B2b corrections on real-time positioning with SF, DF, and MF PPP models,

TABLE III
RECEIVER INFORMATION OF THE MGEX STATIONS

STATIONS	Receivers	Antenna
GAMG	SEPT POLARX5TR	LEIAR25.R4
JFNG	TRIMBLE ALLOY	TRM59800.00
MIZU	SEPT POLARX5	SEPCHOKE_B3E6
PTGG	SEPT POLARX5	TRM59800.00
ULAB	JAVAD TRE_3	JAVRINGANT_G5T
URUM	JAVAD TRE_3	JAVRINGANT_G5T

TABLE IV
PPP STRATEGY

Items	Description
Observables type	BDS-3:B1I/B3I/B1C/B2a
Elevation cut-off	7°
Sampling rate	30s
Estimator	SF-PPP: Least-square Estimator [1]; Other PPP models: Forward Kalman filter.
PCO/PCV	IGS14-2178.atx
Troposphere	GMF model + Random walk
Ambiguity	Float
Receiver clock offset	White noise
Other corrections with model	Solid Earth tide, Ocean tide, Pole tide, Phase wind-up.
Experimental schemes	SF-PPP: B1I using PPP-B2b products; DF-PPP: B1I/B3I IF combination using PPP-B2b products; MF-PPP: B1I/B3I and B1C/B2a IF combinations using PPP-B2b products; Reference: B1I/B3I IF combination with final GFZ products.

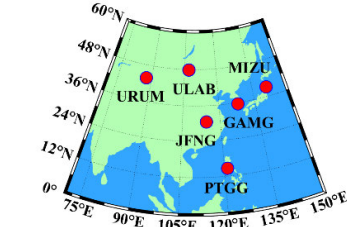


Fig. 4. Distribution of MGEX stations.

positioning experiments were conducted using the BDS-3 observations from six MGEX stations in and around China. The dataset is from June 1 to 30, 2022. The distribution and details of these MGEX stations are shown in Fig. 4 and Table III. The experiment is divided into four schemes, namely, SF PPP, DF PPP, MF PPP, and reference experiments, and each scheme is tested by both static and simulated kinematic solutions. The reference experiments are the DF IF PPP using final GFZ products. The detailed PPP strategy is shown in Table IV.

Considering the difference of the geometry of satellites and the number of available satellites in different observation periods, the PPP solutions were restarted every 4 h, and six positioning results can be obtained per day for each station. Then, each station can get 180 PPP results in 30 days. The reference coordinates of MGEX stations are the IGS weekly solution. The positioning errors of these 180 segments for each station were fused according to each epoch, and the median, 68.26% (1σ) value, and 95.44% (2σ) value of statistical results were given to evaluate the positioning accuracy and

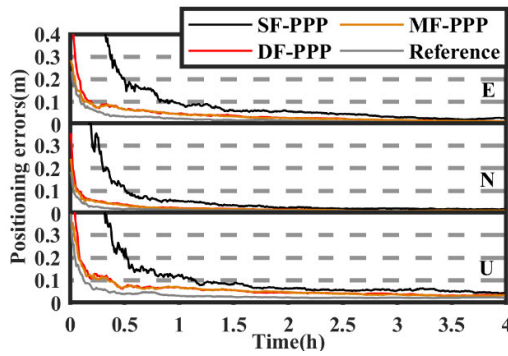


Fig. 5. Median positioning error time series with four PPP models in static solutions for PTGG station.

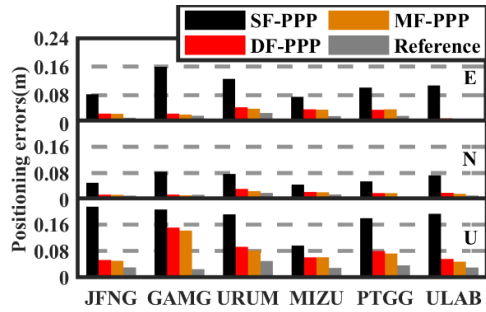


Fig. 6. Median positioning errors of six stations with four PPP models in static solution.

convergence time. The convergence time was defined as the horizontal accuracy and vertical accuracy of 20 and 40 cm, respectively, and lasted for more than 5 min [6].

B. Static Test

In this study, we use the positioning results from PTGG station as an example, and the median positioning error time series in the *ENU* directions for static model are shown in Fig. 5. We interrupt the PPP filter every 4 h and compute the median position error time series using 30 days' data to explore the convergence characteristics of different PPP models. The black line, red line, yellow line, and gray line represent the median positioning errors of the SF PPP, DF PPP, MF PPP, and reference solutions, respectively. The figure shows that the SF PPP converges slowly, and it takes about 30 min to converge to 0.2 m in the *ENU* directions. The accuracy of DF PPP and MF PPP is comparable, and it takes about 10 min to converge to 0.2 m in the *ENU* directions, but the convergence speed of MF PPP in the initial stage is better than that of DF PPP. Compared with the reference results, there is still room for improvement in the positioning accuracy after convergence.

The median of positioning errors and convergence time of the six stations is shown in Figs. 6 and 7. The SF PPP positioning results are relatively poor, except for the *N* direction, the accuracies of the *E* and *U* directions are all greater than 0.1 m. The DF PPP and MF PPP results of all stations are better than 0.1 m in the *ENU* directions except the GAMG station. Among them, the positioning accuracy in the *E* and *N* directions is better, which are 4 and 2 cm, respectively. It shows that the

TABLE V
MEAN OF STATISTICAL RESULTS OF POSITIONING ERRORS WITH FOUR PPP MODELS

PPP model	Static accuracy(m)			Kinematic accuracy(m)		
	Median	1 σ	2 σ	Median	1 σ	2 σ
E	SF-PPP	0.109	0.172	0.428	0.234	0.400
	DF-PPP	0.033	0.050	0.119	0.048	0.079
	MF-PPP	0.031	0.049	0.113	0.046	0.075
	Reference	0.021	0.033	0.080	0.036	0.060
N	SF-PPP	0.064	0.094	0.195	0.145	0.225
	DF-PPP	0.021	0.031	0.061	0.027	0.041
	MF-PPP	0.019	0.028	0.056	0.025	0.037
	Reference	0.014	0.021	0.044	0.019	0.029
U	SF-PPP	0.180	0.271	0.572	0.394	0.601
	DF-PPP	0.082	0.113	0.206	0.110	0.158
	MF-PPP	0.076	0.105	0.194	0.101	0.144
	Reference	0.033	0.051	0.113	0.054	0.082

TABLE VI
MEAN OF STATISTICAL RESULTS OF CONVERGENCE TIME WITH FOUR PPP MODELS

Positioning model	PPP model	Convergence time (min)		
		Median	1 σ	2 σ
Static	SF-PPP	50.42	73.50	184.83
	DF-PPP	9.75	13.25	42.83
	MF-PPP	7.33	11.67	37.17
	Reference	7.58	10.92	26.25
Kinematic	SF-PPP	144.3	226.08	238.00
	DF-PPP	15.67	27.58	107.00
	MF-PPP	11.33	22.08	98.00
	Reference	11.75	20.92	64.92

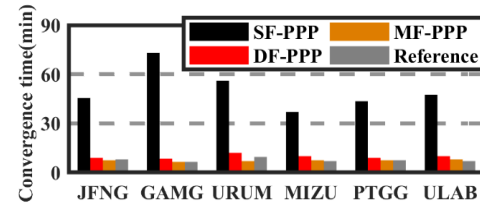


Fig. 7. Median convergence time of six stations with four PPP models in static solution.

improvement effect of DF PPP and MF PPP with PPP-B2b products on the *E* and *N* directions is the most obvious, which is basically consistent with the reference results. Compared with DF PPP, the accuracy of MF PPP is improved by 6.1%, 9.5%, and 7.3% in the *ENU* directions, respectively. As can be seen from Fig. 7, the convergence time of SF PPP is longer, greater than 40 min, and far more than the 10 min of DF PPP and MF PPP. The convergence time of MF PPP is even better than that of reference results at some stations. Compared with DF PPP, the convergence time of MF PPP is improved by 24.8%. It shows that although more frequencies are used, the positioning accuracy is not significantly improved after convergence, but in the initial state, MF PPP can converge faster.

We compared the performance of the four different PPP models in static solution with the following three measures: median, 1σ value, and 2σ value of positioning errors and convergence time. Tables V and VI show the mean values of these performance measures computed from the six stations. With the increase in the statistical range, the positioning accuracy deteriorates, and the convergence time also increases

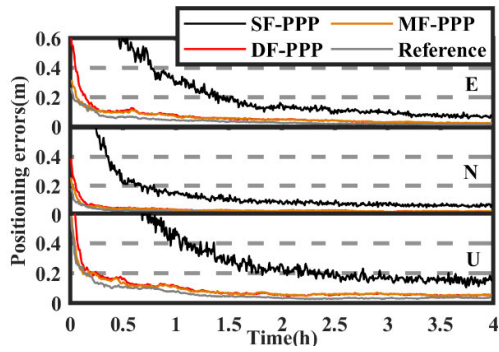


Fig. 8. Median positioning error time series with four PPP models in kinematic solutions for PTGG station.

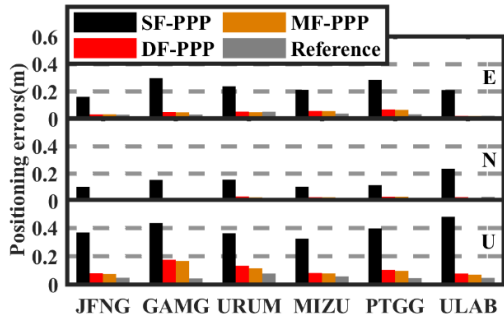


Fig. 9. Median positioning errors of six stations with four PPP models in kinematic solution.

gradually. The 1σ statistical results of DF PPP and MF PPP are better than 0.1 m in the E and N directions and 0.2 m in the U direction, respectively. The 2σ statistical results of DF PPP and MF PPP are better than 0.5 m in the ENU directions. From the perspective of convergence time, the 1σ statistical results of DF PPP and MF PPP are better than 15 min, and the results of MF PPP and reference experiments are more consistent. The mean of matched satellites number for these stations is 8.09–8.61. The available satellites decrease rapidly as the stations get farther away from the PPP-B2b service area. Fewer satellites lead to poorer PPP performance and longer convergence time. Therefore, to ensure the positioning performance of PPP-B2b service to enhance real-time PPP, the service should be applied close to its service area.

C. Kinematic Test

We also use positioning results from PTGG station as an example, and the median positioning error time series in the ENU directions are shown in Fig. 8. The experimental calculation procedure is similar to the static solution. SF PPP takes about 2 h to converge to 0.2 m in the ENU directions. The positioning error time series of DF PPP, MF PPP, and reference experiments are very consistent, and the positioning accuracy after convergence is similar, which is about 15 min to converge to 0.2 m.

In the kinematic model, the median of positioning errors and convergence time of the six stations are shown in Figs. 9 and 10. The accuracy of SF PPP is relatively poor, about 0.2 m in the E and N directions and about 0.4 m in the U direction. After convergence of DF PPP and MF

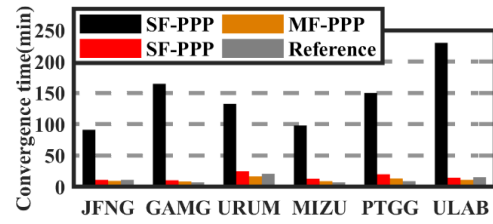


Fig. 10. Median convergence time of six stations with four PPP models in kinematic solution.

PPP, the positioning accuracy in the ENU directions is close to the reference results, and both are better than 0.2 m, especially the N direction is better than 0.04 m. Compared with DF PPP, MF PPP has an increase of 4.2%, 7.4%, and 8.2% in the ENU directions, respectively. From the median convergence time in Fig. 10, it can be seen that SF PPP is more than 90 min, while DF PPP and MF PPP are better than 25 min. Compared with DF PPP, the convergence time of MF PPP is improved by 27.7%. For JFNG, URUM, and ULAB stations, the convergence time of MF PPP is better than that of reference results, which indicates that the increased number of observations contributes to fast convergence.

For the statistical results of six stations with four PPP models in kinematic solution, we also give the median, 1σ value, and 2σ value of positioning errors and convergence time, and Tables V and VI show the mean values of these performance measures. The 1σ statistical results of position errors show that the SF PPP is better than 0.4 m in the E and N directions and 0.6 m in the U direction. DF PPP and MF PPP are better than 0.08 m in the E and N directions and 0.16 m in the U direction. The 2σ statistical results show that the accuracy of SF PPP in the ENU directions is at the meter level, and the accuracy of DF PPP and MF PPP is at the decimeter level. At the same time, the positioning accuracy of DF PPP and MF PPP is close to that of reference results in the E and N directions, but there is still room for improvement in the U direction. In terms of convergence time, the median, 1σ , and 2σ statistical results show that MF PPP is improved by 27.7%, 19.9%, and 8.4% compared with DF PPP, respectively, and the results are more consistent with reference results.

V. CONCLUSION

The new PPP-B2b-enhanced service broadcast by the BDS-3 GEO satellite is freely available to GNSS users in China and its surrounding areas, enabling real-time static centimeter-level and kinematic decimeter-level positioning accuracy. The accuracy of orbit, clock offset, and DCBs of the PPP-B2b products is first evaluated, and then, the augmentation effect of PPP-B2b service on positioning accuracy and convergence time of real-time BDS-3 SF, DF, and MF PPP is analyzed. The PPP-B2b products and the BDS-3 observations of six MGEX stations were used from June 1 to 30, 2022. The relevant conclusions are as follows.

1) The clock offset STD of MEO and IGSO satellites is 0.118 and 0.237 ns, respectively, which is 75.9% and 61.8% higher than that of broadcast ephemeris. The orbit rms of MEO and IGSO satellites in the 3-D direction is 0.286 and

0.447 m, respectively, which are 18.3% and 12.9% higher than that of the broadcast ephemeris. The DCB difference of B1I/B1Cp/B2ap between the PPP-B2b products and the CAS products is within 2 ns, and the DCB of the PPP-B2b products and the TGD has the same accuracy.

2) In static solutions, the statistical results show that the median positioning accuracy of SF PPP, DF PPP, and MF PPP is better than 0.2, 0.09, and 0.08 m, respectively, and the convergence time is better than 51, 10, and 8 min, respectively. Compared with DF PPP, the positioning accuracy of MF PPP is increased by 6.1%, 9.5%, and 7.3% in the *ENU* directions, and the convergence time is increased by 24.8%.

3) In kinematic solutions, the statistical results show that the median positioning accuracy of SF PPP, DF PPP, and MF PPP is better than 0.4, 0.12, and 0.12 m, respectively, and the convergence time is better than 145, 16, and 12 min, respectively. Compared with DF PPP, the positioning accuracy of MF PPP is increased by 4.2%, 7.4%, and 8.2% in the *ENU* directions, and the convergence time is increased by 27.7%.

4) In general, the positioning accuracy of DF PPP and MF PPP with the PPP-B2b products is relatively consistent, close to that of DF PPP using the final precise products. In terms of convergence time, the results of MF PPP are better than that of DF PPP, which is consistent with DF PPP using the final precise products, indicating that the increased observations can effectively accelerate convergence.

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REFERENCES

- [1] C. Zhao, B. Zhang, and X. Zhang, "SUPREME: An open-source single-frequency uncombined precise point positioning software," *GPS Solutions*, vol. 25, no. 3, pp. 1–9, Jul. 2021.
- [2] Y. Yang et al., "Featured services and performance of BDS-3," *Sci. Bull.*, vol. 66, no. 20, pp. 2135–2143, Oct. 2021.
- [3] Z. Zhang, B. Li, L. Nie, C. Wei, S. Jia, and S. Jiang, "Initial assessment of BeiDou-3 global navigation satellite system: Signal quality, RTK and PPP," *GPS Solutions*, vol. 23, no. 4, Oct. 2019.
- [4] H. Ge, B. Li, M. Ge, Y. Shen, and H. Schuh, "Improving BeiDou precise orbit determination using observations of onboard MEO satellite receivers," *J. Geodesy*, vol. 91, no. 12, pp. 1447–1460, Dec. 2017.
- [5] Y. Yang, Q. Ding, W. Gao, J. Li, Y. Xu, and B. Sun, "Principle and performance of BDSBAS and PPP-B2b of BDS-3," *Satell. Navigat.*, vol. 3, no. 1, p. 5, Dec. 2022.
- [6] C. Tang et al., "Orbit determination, clock estimation and performance evaluation of BDS-3 PPP-B2b service," *J. Geodesy*, vol. 96, no. 9, p. 60, 2022.
- [7] M. Li, T. Xu, M. Guan, F. Gao, and N. Jiang, "LEO-constellation-augmented multi-GNSS real-time PPP for rapid re-convergence in harsh environments," *GPS Solutions*, vol. 26, no. 1, pp. 1–12, Jan. 2022.
- [8] A. Allahviridi-Zadeh, K. Wang, and A. El-Mowafy, "POD of small LEO satellites based on precise real-time MADOCA and SBAS-aided PPP corrections," *GPS Solutions*, vol. 25, no. 2, p. 31, Apr. 2021.
- [9] E. Tunali and M. T. Özlüdemir, "GNSS PPP with different troposphere models during severe weather conditions," *GPS Solutions*, vol. 23, no. 3, p. 82, Jul. 2019.
- [10] T. Liu, B. Zhang, Y. Yuan, and X. Zhang, "On the application of the raw-observation-based PPP to global ionosphere VTEC modeling: An advantage demonstration in the multi-frequency and multi-GNSS context," *J. Geodesy*, vol. 94, no. 1, p. 1, Jan. 2020.
- [11] Z. Li, N. Wang, L. Wang, A. Liu, H. Yuan, and K. Zhang, "Regional ionospheric TEC modeling based on a two-layer spherical harmonic approximation for real-time single-frequency PPP," *J. Geodesy*, vol. 93, no. 9, pp. 1659–1671, Sep. 2019.
- [12] X. Li et al., "Multi-constellation GNSS PPP instantaneous ambiguity resolution with precise atmospheric corrections augmentation," *GPS Solutions*, vol. 25, no. 3, p. 107, 2021.
- [13] M. Glaner and R. Weber, "PPP with integer ambiguity resolution for GPS and Galileo using satellite products from different analysis centers," *GPS Solutions*, vol. 25, no. 3, p. 102, Jul. 2021.
- [14] L. Wang and S. Verhagen, "A new ambiguity acceptance test threshold determination method with controllable failure rate," *J. Geodesy*, vol. 89, no. 4, pp. 361–375, Apr. 2015.
- [15] J. Geng, J. Guo, C. Wang, and Q. Zhang, "Satellite antenna phase center errors: Magnified threat to multi-frequency PPP ambiguity resolution," *J. Geodesy*, vol. 95, no. 6, p. 72, Jun. 2021.
- [16] J. Guo, J. Geng, and C. Wang, "Impact of the third frequency GNSS pseudorange and carrier phase observations on rapid PPP convergences," *GPS Solutions*, vol. 25, no. 2, p. 30, Apr. 2021.
- [17] X. Li et al., "Galileo PPP rapid ambiguity resolution with five-frequency observations," *GPS Solutions*, vol. 24, no. 1, p. 24, 2019.
- [18] J. Li, Y. Yang, H. He, and H. Guo, "An analytical study on the carrier-phase linear combinations for triple-frequency GNSS," *J. Geodesy*, vol. 91, pp. 151–166, Feb. 2017.
- [19] V. Duong, K. Harima, S. Choy, D. Laurichesse, and C. Rizos, "An optimal linear combination model to accelerate PPP convergence using multi-frequency multi-GNSS measurements," *GPS Solutions*, vol. 23, no. 2, p. 49, Apr. 2019.
- [20] X. Zhang and P. Li, "Benefits of the third frequency signal on cycle slip correction," *GPS Solutions*, vol. 20, pp. 451–460, Jul. 2015.
- [21] S. Ye et al., "A cycle slip fixing method with GPS + GLONASS observations in real-time kinematic PPP," *GPS Solutions*, vol. 20, no. 1, pp. 101–110, Jan. 2016.
- [22] Z. Xiaohong and L. Xingxing, "Instantaneous re-initialization in real-time kinematic PPP with cycle slip fixing," *GPS Solutions*, vol. 16, no. 3, pp. 315–327, Jul. 2012.
- [23] W. Zhang, Y. Lou, W. Song, W. Sun, X. Zou, and X. Gong, "Initial assessment of BDS-3 precise point positioning service on GEO B2b signal," *Adv. Space Res.*, vol. 69, no. 1, pp. 690–700, Jan. 2022.
- [24] W. Fu, Y. Yang, Q. Zhang, and G. Huang, "Real-time estimation of BDS/GPS high-rate satellite clock offsets using sequential least squares," *Adv. Space Res.*, vol. 62, no. 2, pp. 477–487, Jul. 2018.
- [25] W. Fu, G. Huang, Q. Zhang, S. Gu, M. Ge, and H. Schuh, "Multi-GNSS real-time clock estimation using sequential least square adjustment with online quality control," *J. Geodesy*, vol. 93, no. 7, pp. 963–976, Jul. 2019.
- [26] P. Zhou, H. Yang, G. Xiao, L. Du, and Y. Gao, "Estimation of GPS LNAV based on IGS products for real-time PPP," *GPS Solutions*, vol. 23, no. 1, p. 27, Jan. 2019.
- [27] X. Lu, L. Chen, N. Shen, L. Wang, Z. Jiao, and R. Chen, "Decoding PPP corrections from BDS B2b signals using a software-defined receiver: An initial performance evaluation," *IEEE Sensors J.*, vol. 21, no. 6, pp. 7871–7883, Mar. 2021.
- [28] Y. Xu, Y. Yang, and J. Li, "Performance evaluation of BDS-3 PPP-B2b precise point positioning service," *GPS Solutions*, vol. 25, no. 4, pp. 1–14, Oct. 2021.
- [29] J. Tao, J. Liu, Z. Hu, Q. Zhao, G. Chen, and B. Ju, "Initial assessment of the BDS-3 PPP-B2b RTS compared with the CNES RTS," *GPS Solutions*, vol. 25, no. 4, pp. 1–16, Oct. 2021.
- [30] Z. Ren, H. Gong, J. Peng, C. Tang, X. Huang, and G. Sun, "Performance assessment of real-time precise point positioning using BDS PPP-B2b service signal," *Adv. Space Res.*, vol. 68, no. 8, pp. 3242–3254, Oct. 2021.
- [31] Z. Nie, X. Xu, Z. Wang, and J. Du, "Initial assessment of BDS PPP-B2b service: Precision of orbit and clock corrections, and PPP performance," *Remote Sens.*, vol. 13, no. 11, p. 2050, May 2021.
- [32] Y. Liu, C. Yang, and M. Zhang, "Comprehensive analyses of PPP-B2b performance in China and surrounding areas," *Remote Sens.*, vol. 14, no. 3, p. 643, Jan. 2022.
- [33] H. Yang, S. Ji, D. Weng, Z. Wang, K. He, and W. Chen, "Assessment of the feasibility of PPP-B2b service for real-time coseismic displacement retrieval," *Remote Sens.*, vol. 13, no. 24, p. 5011, Dec. 2021.
- [34] O. Montenbruck, P. Steigenberger, and A. Hauschild, "Broadcast versus precise ephemerides: A multi-GNSS perspective," *GPS Solutions*, vol. 19, no. 2, pp. 321–333, 2015.
- [35] H. Zhou et al., "Impact of higher-order ionospheric delay on the reliability of RTK ambiguity estimation," *Adv. Space Res.*, vol. 69, no. 1, pp. 727–736, Jan. 2022.

- [36] F. Guo, X. Zhang, and J. Wang, "Timing group delay and differential code bias corrections for BeiDou positioning," *J. Geodesy*, vol. 89, no. 5, pp. 427–445, May 2015.
- [37] X. Lin, L. Baojun, L. Yingchun, X. Sujie, and B. Tao, "Satellite geometry and attitude mode of BDS-3 MEO satellites developed by SECM," in *Proc. 31st Int. Tech. Meeting Satell. Division Inst. Navigat.*, 2018, pp. 1268–1289.
- [38] Y. Yang et al., "BeiDou-3 broadcast clock estimation by integration of observations of regional tracking stations and inter-satellite links," *GPS Solutions*, vol. 25, no. 2, p. 57, Apr. 2021.
- [39] N. Wang, Y. Yuan, Z. Li, O. Montenbruck, and B. Tan, "Determination of differential code biases with multi-GNSS observations," *J. Geodesy*, vol. 90, no. 3, pp. 209–228, 2016.



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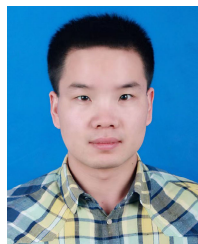


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